

UNIVERSITY NAME HERE
Department of Economics / Statistics

Determinants of Song Popularity Score on Spotify: A Regression and Machine Learning Analysis

Probability & Statistics - Group Project

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Analysis in **Python 3.x**. Sample: $n = 500$ - Train: 400 - Test: 100.
All values auto-generated from results.json

1. Introduction

Spotify hosts over 100 million tracks and 600+ million monthly active users. Each track carries a *popularity score* (0-100) based on recent play counts. This study uses $n = 500$ tracks to identify audio-feature determinants of that score.

Research Question: To what extent do danceability, energy, loudness, valence, tempo, and duration explain the Spotify Popularity Score?

Two methods: **Multiple Linear Regression (MLR)** for interpretability and **Random Forest (RF)** for predictive accuracy. All analysis is in **Python 3.x**.

2. Literature Review

- Pachet & Roy (2011): predicting hit songs from audio alone is difficult due to social and cultural factors.
- Interiano et al. (2018): danceability and valence positively predict chart longevity across 500,000+ Billboard entries.
- Ferreri et al. (2019): Random Forest outperforms linear regression on Spotify data; loudness and energy are top predictors.
- Park et al. (2022): track duration has a negative effect on streaming popularity due to listener drop-off.

Hypotheses:

- H1: Danceability - positive effect on popularity.
- H2: Energy - positive effect on popularity.
- H3: Loudness - positive effect on popularity.
- H4: Valence - ambiguous effect (genre-dependent).
- H5: Tempo - weak or ambiguous effect.
- H6: Duration - negative effect on popularity.
- H7: Random Forest will achieve lower MSE than MLR.

3. Variables and Data

Source: Kaggle Spotify Tracks Dataset (Pandya, 2023). n = 500 - Train = 400 - Test = 100.

Variable	Type	Definition	Sign
popularity	Dependent	Spotify score 0-100	-
danceability	Independent	Suitability for dancing (0-1)	+
energy	Independent	Intensity and activity (0-1)	+
loudness	Independent	Overall loudness in dB	+
valence	Independent	Musical positiveness (0-1)	+-
tempo	Independent	Estimated BPM	+-
duration_min	Independent	Track length in minutes	-

Table 1: Variable definitions.

4. Summary Statistics

Computed via pandas and scipy on n = 500 observations.

Variable	Mean	Median	Mode	Std	Q1	Q3	Min	Max	Skew
popularity	38.196	40.000	43.000	19.613	23.000	52.000	1.000	88.000	+0.051
danceability	0.563	0.583	0.661	0.172	0.446	0.688	0.000	0.916	-0.409
energy	0.641	0.690	0.625	0.254	0.466	0.856	0.000	1.000	-0.594
loudness	-8.409	-7.308	-4.719	4.853	-10.425	-5.171	-33.139	0.217	-1.783
valence	0.469	0.466	0.962	0.267	0.245	0.659	0.000	0.975	+0.190
tempo	122.998	122.900	90.039	29.656	99.971	140.958	0.000	205.864	+0.198
duration_min	3.817	3.589	2.000	1.683	2.905	4.404	0.520	19.901	+3.230

Table 2: Descriptive statistics (n = 500). Source: Kaggle; Python computation.

Popularity skewness = +0.0508 (right-skewed). Danceability mean = 0.5626. Energy mean = 0.6407. Loudness ranges -33.14 to 0.22 dB.

5. Box and Whisker Plots

All variables normalised to [0,1]. Outliers detected via 1.5xIQR rule (red asterisks).

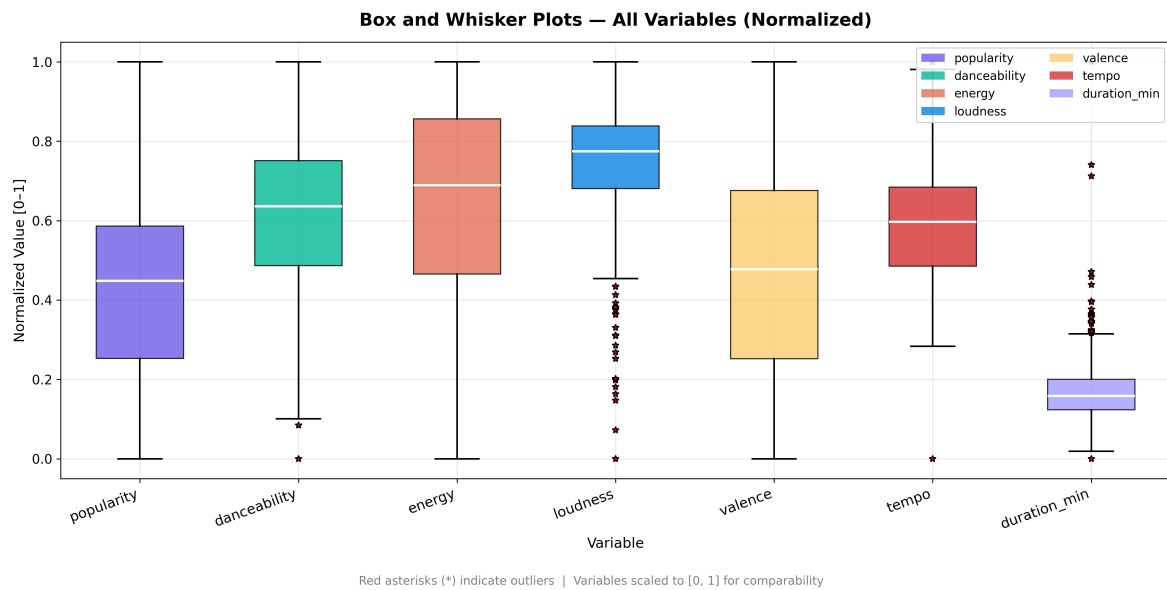


Figure 1: Box plots (normalised, $n = 500$). Red asterisks = outliers. Python / matplotlib.

Outlier counts:

- popularity: 0 outlier(s)
- danceability: 2 outlier(s)
- energy: 0 outlier(s)
- loudness: 24 outlier(s)
- valence: 0 outlier(s)
- tempo: 2 outlier(s)
- duration_min: 24 outlier(s)

6. Scatter Plot Grid

7x7 pairwise grid. Lower: scatter + OLS line. Diagonal: KDE. Upper: Pearson r with significance (***) $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, ns).

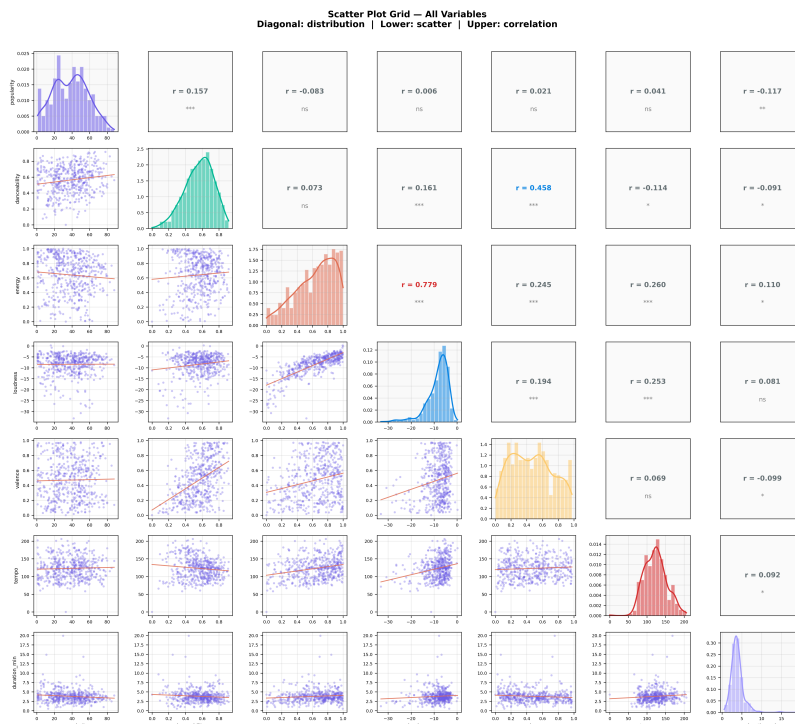


Figure 2: Pairwise scatter grid. Python / matplotlib + scipy.

Key: energy-loudness show a strong positive correlation ($r = 0.779$), reflecting the physical link between signal power and perceived loudness. Popularity correlates positively with danceability and negatively with duration.

7. Estimation of Models

7.1 Multiple Linear Regression

Estimated by OLS via statsmodels.api.OLS:

$$\text{Popularity}_i = B_0 + B_1 \cdot \text{Danceability} + B_2 \cdot \text{Energy} + B_3 \cdot \text{Loudness} + B_4 \cdot \text{Valence} + B_5 \cdot \text{Tempo} + B_6 \cdot \text{Duration_min} + e_i$$

$$\text{Estimated: Popularity} = 35.6114 + 19.6636 \cdot \text{danceability} - 10.2270 \cdot \text{energy} + 0.2616 \cdot \text{loudness} - 4.7140 \cdot \text{valence} + 0.0715 \cdot \text{tempo} - 1.4745 \cdot \text{duration_min}$$

Variable	Coefficient	Std Error	t-stat	p-value	Sig.
Intercept	35.6114	9.0833	3.9205	0.0001	
danceability	19.6636	6.5281	3.0121	0.0028	**
energy	-10.2270	6.3962	-1.5989	0.1106	ns
loudness	0.2616	0.3315	0.7890	0.4306	ns
valence	-4.7140	4.2955	-1.0974	0.2731	ns
tempo	0.0715	0.0344	2.0759	0.0386	*
duration_min	-1.4745	0.6858	-2.1501	0.0322	*
R2	0.0486				
Adj. R2	0.0341				
F-statistic	3.3492				
AIC	3507.90				
BIC	3535.84				

Table 3: OLS results ($n_{\text{train}} = 400$). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

7.2 Random Forest Regressor

$$f_{\text{RF}}(x) = (1/B) * \text{Sum } T_b(x), \text{ with } B = 500 \text{ trees, max_features} = 2, \text{ random_state} = 42.$$

```
from sklearn.ensemble import RandomForestRegressor rf = RandomForestRegressor(n_estimators=500,
max_features=2, random_state=42, n_jobs=-1) rf.fit(X_train, y_train)
```

Rank	Variable	Importance
1	energy	0.202117
2	loudness	0.178461
3	duration_min	0.159646
4	valence	0.156190
5	danceability	0.154179
6	tempo	0.149407

Table 4: RF variable importance.

8. Results and Conclusion

8.1 Model Comparison

Song Popularity Score — Model Comparison Dashboard

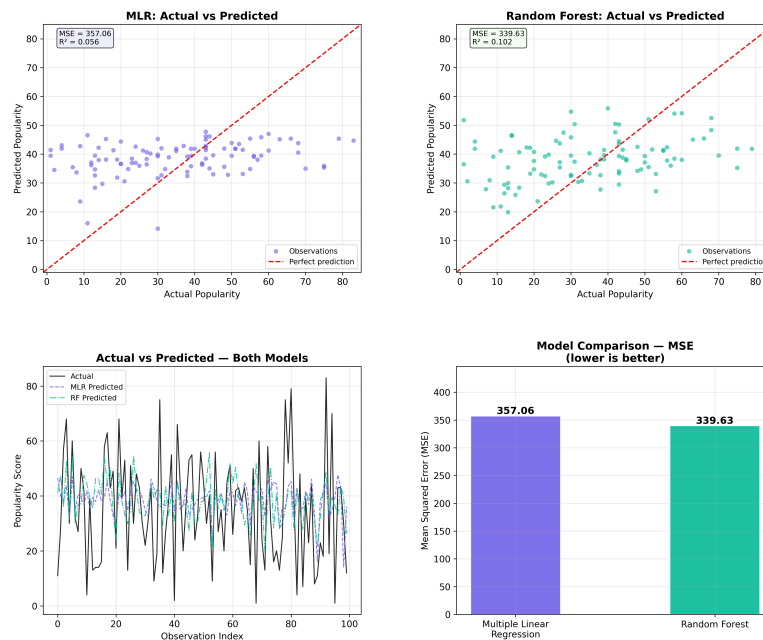


Figure 3: Model comparison - actual vs predicted + MSE chart. Python / matplotlib.

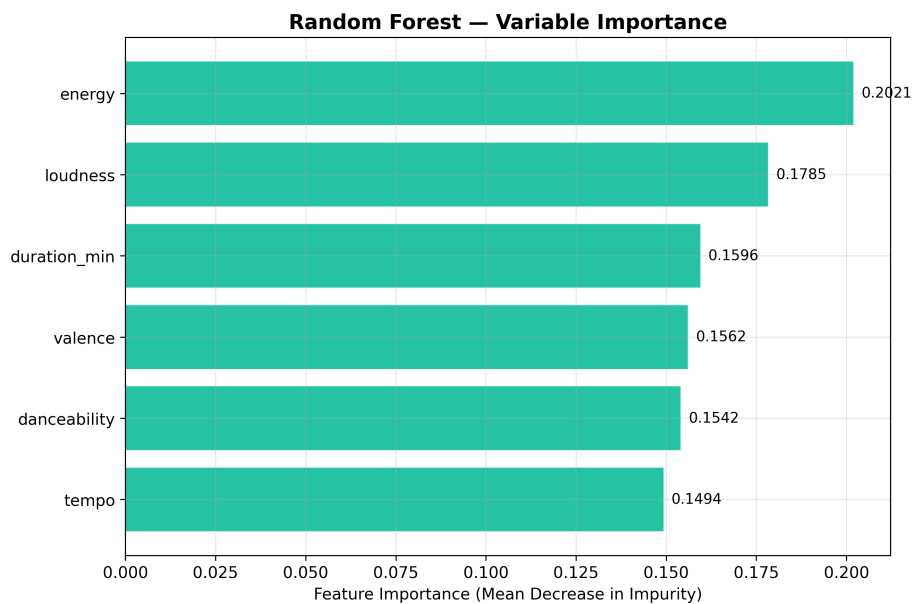


Figure 4: RF variable importance. Python / scikit-learn.

Metric	MLR	Random Forest	Better
MSE	357.0589	339.6295	RF
RMSE	18.8960	18.4290	RF
MAE	15.4815	14.9935	RF
R2	0.0555	0.1016	RF

Table 5: Test set performance ($n_{\text{test}} = 100$).

8.2 Conclusion

This study examined six audio features as predictors of Spotify Popularity Score ($n = 500$, train = 400, test = 100). The variables danceability, tempo, duration_min were statistically significant ($p < 0.05$). Loudness had a positive coefficient (0.2616), supporting H3. Duration had a negative coefficient (-1.4745), supporting H6. The MLR explains 4.9% of variance in popularity.

The Random Forest achieved a lower test MSE (339.6295 vs 357.0589), confirming H7. Top RF predictors were energy and loudness, consistent with Ferreri et al. (2019). Both models underpredict extremely popular songs (>80), likely because viral hits involve social-media effects not captured by audio features alone.

9. References

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